

DBDOME Setup

Installation · Deploy · Runscript · Signing

High-Level Architecture & Process

dbdome_setup.exe (full install) · dbdome_update.exe (incremental) · deploy scripts (live redeploy)

Payload: DBDOME appliance + Grafana 12.2.0 + PostgreSQL 18.3 + ODBC + Oracle client + DB seed + migrations

Authenticode-signed binaries · idempotent SQL runscript · service-based runtime on C:\ProgramData\DBDOME

by dbexpert.ai · sources: dbdome_setup/setup.py · deploy_*.ps1 · sql_script_runner.py · sign_dbdome.ps1

Three delivery artifacts

- **dbdome_setup.exe** — full install (v2.0.0) — clean-host installer (PyInstaller, setup.spec): installs PostgreSQL 18.3, ODBC & Oracle client, restores the DB, registers services, configures Grafana & firewall. Run as Administrator.
- **dbdome_update.exe** — incremental updater — same payload shape; refreshes appliance binaries + newer postgres\install migrations on an existing install (no PG reinstall).
- **deploy_*.ps1** — developer redeploy — **deploy_redeploy.ps1** (live only) and **deploy_all_three.ps1** (stage both package payloads + live) swap **_internal** + the two exes into **C:\ProgramData\DBDOME\bin**.
- Everything DBDOME-authored is Authenticode-signed — **sign_dbdome.ps1** signs the exes + installers; the public **.cer** is imported into the host trust store at install time.
- Schema applied at runtime, not by the installer — the service runs **sql_script_runner** on startup + daily to apply ordered, checksum-tracked SQL migrations idempotently.

Setup package composition

dbdome_setup.exe
installer (setup.py)

dbdome\ payload

copied to
C:\ProgramData\DBDOME

bin

dbdome_dbanalytics.exe · dbdome_service.exe · _internal · certification\dbdome_cert.cer · nssm.exe · .env

Grafana 12.2.0

dbdomeg-server/cli.exe · conf\ · data\grafana.db · public\ · plugins-bundled\

postgres\install

operator migrations 62xx-73xx · dbanalytics_install.backup (pg_restore seed)

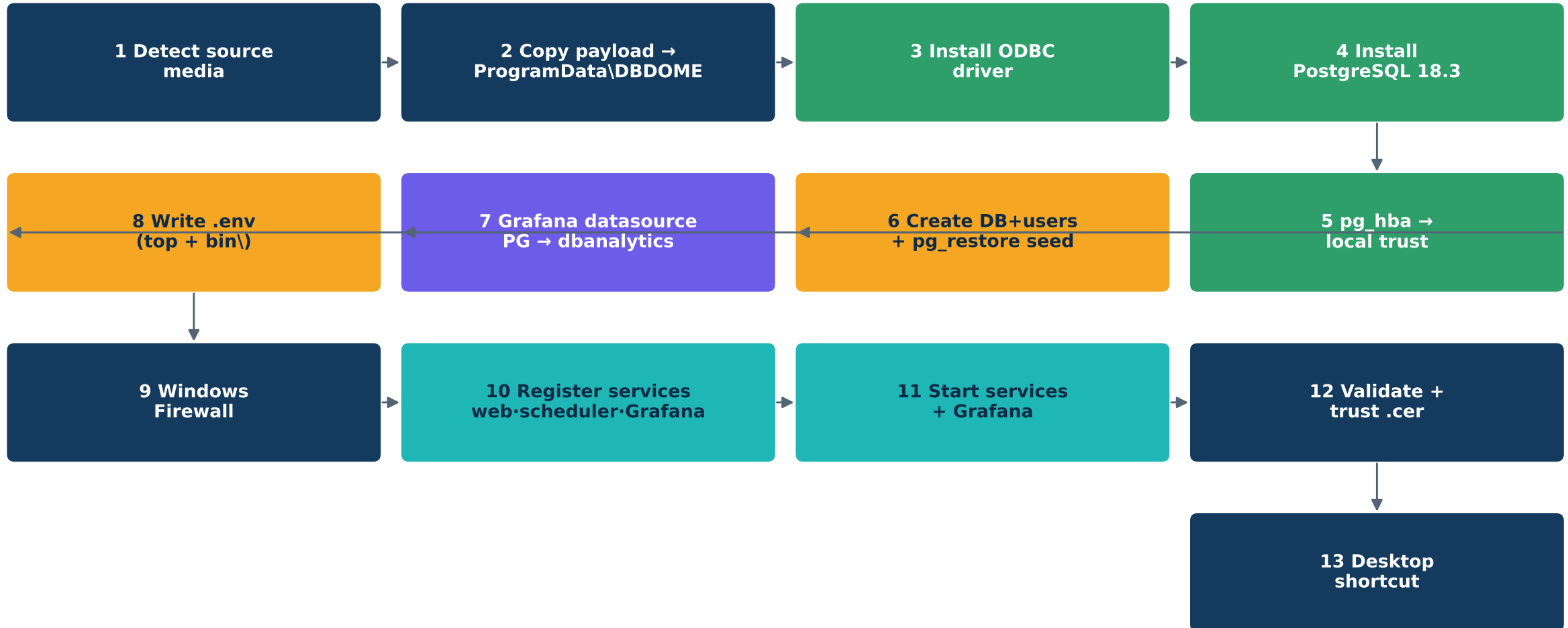
bundled installers

postgresql-18.3-windows-x64.exe · msodbcsql.msi · instantclient_23_0 (Oracle)

scripts\ · VERSION

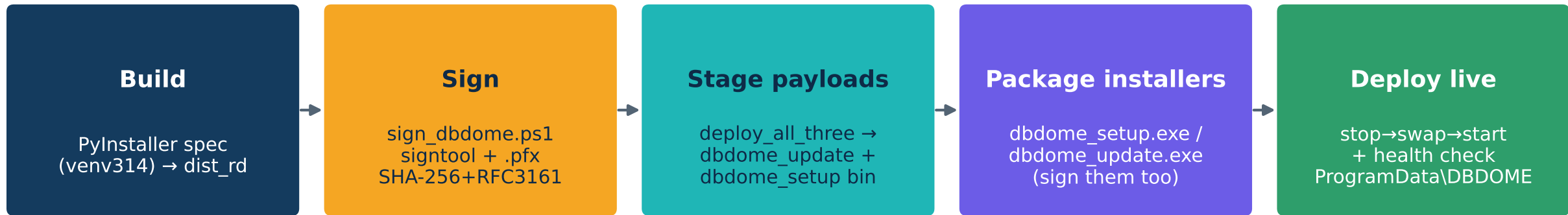
dbanalytics schema SQL · version stamp

Installation flow — dbdome_setup.exe



Installs a dedicated PostgreSQL 18.3 (local trust via pg_hba), DB dbanalytics with users dbdome_adm / dbdome_mon_usr, restores dbanalytics_install.backup, and registers DBDOME_web + DBDOME_scheduler (dbdome_service.exe) and DBDOME_Grafana (NSSM).

Build → Sign → Package → Deploy



- **Signed once, trusted everywhere** — the same signed dist_rd feeds the live install and both package payloads; customers receive already-signed binaries and the shipped .cer makes them trusted (no “Unknown Publisher”).
- **Schema follows separately** — new SQL migrations are dropped into postgres\install (shipped) and applied by the runtime runscript — no rebuild to ship a detection/view change.

Deploy process

- **Targets** — `deploy_all_three.ps1` → (1) `dbdome_update\dbdome\bin (stage)` · (2) `dbdome_setup\dbdome\bin (stage)` · (3) `C:\ProgramData\DBDOME\bin (LIVE)`. `deploy_redeploy.ps1` = live only.
- **Minimal swap** — only `_internal` + `dbdome_dbanalytics.exe` + `dbdome_service.exe` are replaced. `bin\.env` and third-party binaries (`grafana/dbdomeg`, `nssm`, `testers`) are never touched.
- **Sequence (live)** — Stop `DBDOME_scheduler` + `DBDOME_web` → wait for `dbdome_service.exe` to release file locks (abort if still running) → back up (`_internal.bak_<ts>`, `*.bak_<ts>`) → `robocopy /E new _internal` → copy new exes → Start services.
- **Health gate** — `Assert-DeployHealthy` polls ~60s for both services Running AND :8080 listening; warns + points to `dbdome_service.log` if a build crashed on startup.
- **Safety** — `finally-guard` always restarts services even if the swap threw; `robocopy exit ≥ 8` = failure (restore the `.bak`); full rollback recipe printed at the end.

Runscript — sql_script_runner.py

- **Ordering** — every *.sql applied in strict case-insensitive FILENAME order; a zero-padded sequence prefix (e.g. 6690_, 7340_) drives order.
- **Run-once vs re-apply** — each script is checksummed (line-ending-insensitive SHA-256); runs only when its (filename, checksum) hasn't succeeded before. One-shot migration runs once; edited CREATE OR REPLACE re-applies unchanged = cheap no-op.
- **Ledger** — meta.schema_migrations (auto-created): filename, checksum, applied_at, execution_ms, success, error.
- **Concurrency & atomicity** — pg_advisory_lock(727274) serialises concurrent runners (every service startup calls it); each script + its ledger row commit in one transaction.
- **Fail-open** — a failing script is logged + recorded, then the run CONTINUES; one broken view never blocks the rest.
- **Folder priority (additive, first-wins)** — SQL_SCRIPTS_DIR → ../postgres/install → exe/postgres/install → exe/sql_scripts → bundled_internal/sql_scripts → repo. Shipped + operator migrations feed one ledger.
- **When** — invoked at each service startup and on a daily schedule.

Signature — Authenticode signing & trust

- **Certificate** — `certification.py` generates a self-signed RSA-2048 / SHA-256 code-signing cert (CN=dbdome) → `dbdome_cert.{cer,pem,pfx}`.
- **Signing** — `sign_dbdome.ps1` uses the newest Windows-SDK signtool: `sign /f dbdome_cert.pfx /tr <RFC3161 timestamp /td SHA256 /fd SHA256` — SHA-256 digest + trusted timestamp (survives cert expiry).
- **What gets signed** — fresh `dist_rd` build exes, the `dbdome_update` and `dbdome_setup` payload exes, AND the two installer executables (`dbdome_setup.exe` / `dbdome_update.exe`).
- **What is NOT signed** — third-party binaries (`grafana/dbdomeg`, `nssm`) keep their vendors' signatures; never re-signed.
- **Trust on the target** — the public `dbdome_cert.cer` ships in `bin\certification` and the installer imports it into the machine trust stores, so signed DBDOME binaries run without an “Unknown Publisher” prompt.
- **Verify** — `signtool verify /pa` confirms the chain; live exes are signed by redeploying the signed `dist_rd` (they're locked while services run).

Runtime after install



- Two DBDOME services from one exe — `dbdome_service.exe` registers **DBDOME_web** and **DBDOME_scheduler**; **DBDOME_Grafana** is `dbdomeg-server.exe` wrapped by NSSM.
- State — all in the local PostgreSQL 18.3 (`dbanalytics`); secrets in `bin\.env` (read by the exes next to them); migrations reconciled every startup.

Targets & artifacts matrix

Location	Role	Services?	Updated by
C:\dev\...\dist_rd\dbdome_dbanalytics	fresh PyInstaller build	no	build + sign_dbdome.ps1
C:\installs\dbdome_setup\dbdome	full-install payload (+PG/ODBC/Oracle/backup)	no	deploy_all_three (stage)
C:\installs\dbdome_update\dbdome	incremental-update payload	no	deploy_all_three (stage)
C:\ProgramData\DBDOME\bin	LIVE runtime	web · scheduler · Grafana	deploy_redeploy / all_three
PostgreSQL 18.3 · dbanalytics	state store + catalog	(PG service)	installer restore + runscript

- **Golden rule — binaries flow build → sign → (stage payloads + live); schema flows postgres\install (+bundled) → runscript → meta.schema_migrations. bin\.env is never overwritten by deploy.**

Operational safety, rollback & key files

- **Deploy safety** — abort if dbdome_service.exe still holds file locks; per-target timestamped backups; robocopy exit-code check; health gate on services + :8080; finally-guard restart; printed rollback recipe.
- **Runscript safety** — advisory-locked single runner; per-script transaction atomic with its ledger row; fail-and-continue; unchanged scripts skipped.
- **Signing safety** — trusted RFC3161 timestamp (validity survives cert expiry); third-party binaries never re-signed; .cer trust import scoped to the machine store.
- **Key files** — dbdome_setup/setup.py (13-step installer) · deploy_redeploy.ps1 · deploy_all_three.ps1 · processes/sql_script_runner.py · certification/{certification.py, sign_dbdome.ps1, dbdome_cert.*} · payload postgres\install\ (migrations + dbanalytics_install.backup).
- **Secrets** — DB & PFX passwords live only in the operator scripts / host .env — never in the shipped catalog or this document.